

FUS Planner + FUS Simulator: Guide

There are two programs:

Program	What it does
fus_planner	Look at a mouse brain atlas, pick a target region, and lay out where the ultrasound spots should go. Exports a plan as a CSV file.
fus_simulator	Simulate the actual ultrasound beam through the mouse skull, calculate pressure and heating, and check a planner CSV for overlapping hot spots.

You normally use the planner first, then feed its CSV into the simulator. Or, if you just want to look at pressure and heating for your parameters, use the simulator directly.

Part 1: Install the software on windows

Skip to Part 1b if you have a Mac.

Step 1.1 Install Python 3.12

Download Python **3.12** from <https://www.python.org/downloads/release/python-3129/> (scroll to the bottom and pick **Windows installer (64-bit)**).

IMPORTANT: Do not use Python 3.13 or 3.14. The simulator's solver does not work on them. Python 3.12 is the tested version.

On the very first screen of the installer, **tick the box that says "Add python.exe to PATH"**, then click *Install Now*.

Check it worked. Open PowerShell (press **Start**, type **powershell**, press Enter) and run:

```
python --version
```

It must print **Python 3.12.x**. If it prints something else or says the command is not found, restart your computer and try again.

Step 1.2 Install Git

Download and install Git from <https://git-scm.com/download/win>. Accept all the default options.

Check it worked:

```
git --version
```

Step 1.3 Download the code

Pick a folder to keep the project in. This guide uses your Documents folder. When you pick a folder you can right click onto it and you should see 'Information'. There you can see the full path of the folder. Copy it and then paste it in powershell these lines one by one:

- cd \$HOME\Documents
- git clone <https://github.com/valeriadosso25520-dev/In-vivo-tFUS-planner.git>
- cd In-vivo-tFUS-planner

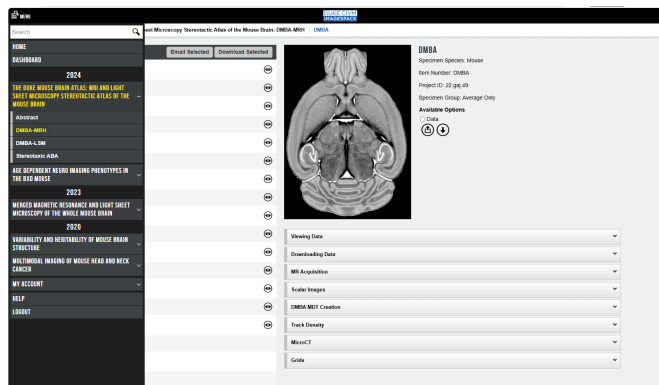
You are now inside the project folder. **Every command from here on assumes you are in this folder.** If you ever get lost, run **cd \$HOME\Documents\In-vivo-tFUS-planner**.

Step 1.4 Add the mouse data

The mouse scans are far too large for GitHub (about 9 GB), so they are **not** included in the download. Compile them all inside a folder (e.g Mouse DUKE).

Go here: [The Duke Mouse Brain Atlas: MRI and light sheet microscopy stereotaxic atlas of the mouse brain | Science Advances](https://www.sciencedirect.com/science/article/pii/S0969216220300000) at the bottom there is a link for the dataset. Request access.

Then once in download these files:



Objects		Email Selected	Download Selected
☐	DMBA ad	☐	☐
☒	DMBA dwi	☐	☐
☐	DMBA nqa	☐	☐
☐	DMBA rd	☐	☐
☐	DMBA md	☐	☐
☐	DMBA gqi-color	☐	☐
☐	DMBA fa	☐	☐
☐	DMBA m0	☐	☐
☐	DMBA m1	☐	☐
☐	DMBA m2	☐	☐
☐	DMBA m3	☐	☐
☐	DMBA tdi5um-color	☐	☐
☐	DMBA iso	☐	☐
☐	DMBA mGRE-unmasked	☐	☐
☒	DMBA CT-cropped	☐	☐
☒	DMBA_RCCF_labels	☐	☐
☐	DMBA_xy_grid	☐	☐
☐	DMBA_xz_grid	☐	☐
☐	DMBA_yz_grid	☐	☐

Every `.nhdr` file needs its matching `.raw` file next to it. The `.nhdr` is just a small text header; the `.raw` holds the actual image and is the big one. If you accidentally delete one, the program will fail to load it.

Step 1.5 Set up fus_planner

```
PS C:\Users\vd224\Documents> cd In-vivo-tFUS-planner
PS C:\Users\vd224\Documents\In-vivo-tFUS-planner> cd fus_planner
PS C:\Users\vd224\Documents\In-vivo-tFUS-planner\fus_planner> |
```

Open powershell and you should be inside the in vivo tfus planner (look at the path and paste it after 'cd'). Then paste:

```
cd fus_planner
```

```
python -m venv .venv
```

```
.\venv\Scripts\pip install --upgrade pip
```

```
.\venv\Scripts\pip install --no-cache-dir -r requirements.txt
```

```
cd ..
```

Step 1.6 Set up fus_simulator

```
cd fus_simulator
```

```
python -m venv .venv
```

```
.\venv\Scripts\pip install --upgrade pip
```

```
.\venv\Scripts\pip install --no-cache-dir -r requirements.txt
```

```
cd ..
```

Part 1b: Install the software on Mac

Step 1b.0 Check which Mac you have

Open Terminal (press Cmd+Space, type Terminal, press Enter) and run:

```
uname -m
```

arm64 means Apple Silicon (M1/M2/M3/M4). **x86_64** means an Intel Mac. Note which one; it matters for the simulator's solver.

Step 1b.1 Install Python 3.12

On Mac, use conda.

If your Terminal prompt already starts with **(base)**, you have conda. If not, install Miniconda from <https://docs.conda.io/en/latest/miniconda.html>, choosing the macOS installer that matches your chip from Step 1b.0.

Then create one Python 3.12 environment per program:

```
conda create -n fus_planner python=3.12 -y
conda create -n fus_simulator python=3.12 -y
```

Step 1b.2 Install Git

Check whether you already have it:

```
git --version
```

If it is missing, macOS will offer to install the developer command line tools. Accept, wait for it to finish, then run the check again.

Step 1b.3 Download the code

```
cd ~/Documents
git clone https://github.com/valeriadosso25520-dev/In-vivo-tFUS-planner.git
cd In-vivo-tFUS-planner
```

Every command from here assumes you are inside this folder. If you get lost, run **cd ~/Documents/In-vivo-tFUS-planner**. Note that Mac paths use forward slashes and **~/** instead of **\$HOME** with backslashes.

Step 1b.4 Add the mouse data

Exactly as in Step 1.4: the ~9 GB dataset is not included and must be requested from the Duke atlas link. Every `.nhdr` needs its matching `.raw` beside it. You do not have to place the folder inside the project; you can leave it anywhere on your Mac and point the program at it using the file boxes in the Data tab.

Step 1b.5 Set up fus_planner

```
conda activate fus_planner
cd ~/Documents/In-vivo-tFUS-planner/fus_planner
pip install --upgrade pip
pip install --no-cache-dir -r requirements.txt
cd ..
```

Step 1b.6 Set up fus_simulator

```
conda activate fus_simulator
cd ~/Documents/In-vivo-tFUS-planner/fus_simulator
pip install --upgrade pip
pip install --no-cache-dir -r requirements.txt
cd ..
```

The first time you run an actual simulation, k-Wave downloads its solver binaries automatically. This needs internet access at that moment. On Apple Silicon this works normally; on an Intel Mac the C++ solver may not be available.

Step 1b.7 Starting the programs on Mac

Wherever the rest of this guide says `\\.venv\\Scripts\\python run_gui.py`, on Mac you instead activate the environment and use a plain `python`:

```
conda activate fus_planner
cd ~/Documents/In-vivo-tFUS-planner/fus_planner
python run_gui.py
```

Or

```
conda activate fus_simulator
cd ~/Documents/In-vivo-tFUS-planner/fus_simulator
python run_gui.py
```

Leave the Terminal window open while the program runs; closing it closes the program. If the window opens behind your other windows, click the Python icon in the Dock to bring it forward.

Part 2: Using fus_planner

Start it

```
cd fus_planner
```

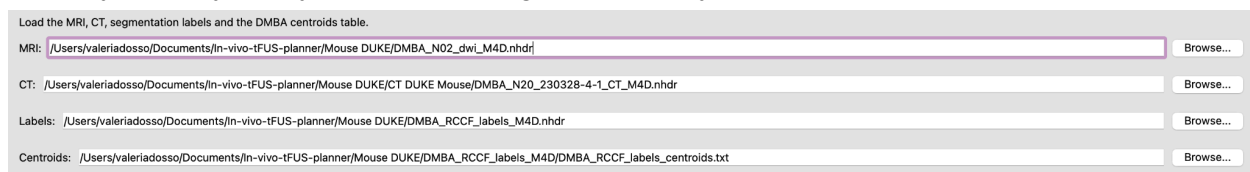
```
..\venv\Scripts\python run_gui.py
```

Leave the PowerShell window open while the program runs. Closing it closes the program.

Remember for Mac you need the code in **1b.7**

Tab 1: Data

1. Click **Use bundled paths**. This fills in all four file boxes automatically. Use the non cropped skull.
2. Click **Load**. This takes a while, the files are large. Wait for the status line.
3. Click **Save as default** (the red button). Now the paths are remembered and filled in for you every time you open the program. You only ever do this once.



The screenshot shows a GUI window titled "Load the MRI, CT, segmentation labels and the DMBA centroids table." It contains four rows of input fields, each with a "Browse..." button to its right. The first row is for "MRI:" with the path "/Users/valeriadosso/Documents/in-vivo-tFUS-planner/Mouse DUKE/DMBA_N02_dwi_M4D.nhdr". The second row is for "CT:" with the path "/Users/valeriadosso/Documents/in-vivo-tFUS-planner/Mouse DUKE/CT DUKE Mouse/DMBA_N20_230328-4-1_CT_M4D.nhdr". The third row is for "Labels:" with the path "/Users/valeriadosso/Documents/in-vivo-tFUS-planner/Mouse DUKE/DMBA_RCCF_labels_M4D.nhdr". The fourth row is for "Centroids:" with the path "/Users/valeriadosso/Documents/in-vivo-tFUS-planner/Mouse DUKE/DMBA_RCCF_labels_M4D/DMBA_RCCF_labels_centroids.txt".

Tab 2: Plan

1. **Set the landmarks.** Use the picker image at the bottom left. Choose **lambda** in the *Picking* dropdown and click on the lambda suture in the image. Then choose **left_eye** and click the bottom of the left eye. Switch the *View* dropdown to **axial slice (heatmap)** and use the slider if you need to look at different depths.
2. **Set the transducer numbers** (top right): the focal spot size (FWHM).
3. **Choose what to target** (bottom right): search for a brain region and tick it, or pick a whole-brain scenario.
4. **Choose a strategy.**
 - **Centroid.** The beam is centred on the centroid of the selected brain region. If the region is larger than a single focal spot, additional target points are added around the centroid to extend coverage.
 - **Coverage.** Target points are placed to maximise the volume of the selected region that falls within the focal volume, rather than anchoring to the centroid. Use this when the region is irregularly shaped, or when the centroid sits outside the region itself.

- **Max coverage.** The region is packed with as many non-overlapping focal spots as will fit, giving the broadest possible coverage at the cost of a longer sonication protocol.
- **Overlap threshold.** All three strategies use a threshold that controls how selective the placement is. It sets the minimum fraction of a candidate focal volume that must fall inside the target region for that point to be accepted. A threshold of 15% means a point is only kept if at least 15% of its focal volume lies within the region of interest.
- Lower threshold (e.g. 15%): more permissive. More target points, broader coverage of the region, but more energy deposited in surrounding tissue.
- Higher threshold (e.g. 60%): more conservative. Fewer target points, tighter confinement to the region, but parts of the region may be left uncovered.

5. Click **Run plan**.

Tab 3: Results

You get three views (xy, xz, yz) with the spots drawn on the brain, plus a table listing every spot with its coordinates relative to lambda and the eye.

- **Save CSV** writes the plan file. **This is the file the simulator reads.** Save it somewhere.
- **Save PDF** writes a printable report of the same thing.

Part 3: Using fus_simulator

Start it

```
cd fus_simulator
```

```
..\venv\Scripts\python run_gui.py
```

Tab 1: Settings (do this once)

1. **CT / prepared skull**: browse to `Mouse DUKE\CT DUKE Mouse\DMBA_N20_230328-4-1_CT-cropped_M4D.nhdr` (use the **cropped** one here).
2. **Atlas labels (centroids)**: browse to `Mouse DUKE\DMBA_RCCF_labels_M4D\DMBA_RCCF_labels_centroids.txt`
3. Click **Save as default** (red button).

From now on both are filled in automatically when you open the program.

Tab 2: Parameters

1. **Transducer settings**: frequency, aperture diameter, focal length. If your transducer has a hole in the middle, tick *Model a central hole*.
2. **Skull**: click **Prepare / Load** (red button). The first time, this reads the CT and builds a skull file. It takes a few seconds and shows a progress bar. Afterwards it loads instantly.
3. You should now see the mouse skull with the transducer drawn next to it. The red star is the focus.
4. **To aim at a brain region** (optional): type in the search box, for example `hippocampus`, click a region in the list, and press **Target selected region**. The skull moves so that region sits exactly on the focus. If you would rather aim by hand, ignore this and use the **AX / LR / AP** offset boxes instead.

Tab 3: Simulation

Numerics, in plain terms:

Setting	What it means	Sensible value
Points per wavelength (PPW)	How fine the calculation grid is. Higher = slower, much more memory, but also more accurate.	At least 15 (1 PPW at thinnest part of the skull assumed to be 0.2 mm)

Setting	What it means	Sensible value
CFL number	Time-step safety. Lower = slower but more stable.	0.1 to 0.3
Tone-burst cycles	How long the pulse is. Don't change this if you want to look at the steady-state pressure.	60

Min Gon Kim, Chih-Yu Yeh, Kai Yu, Zherui Li, Kalpna Gupta, Bin He; Analgesic effect of simultaneously targeting multiple pain processing brain circuits in an aged humanized mouse model of chronic pain by transcranial focused ultrasound. APL Bioeng. 1 March 2025; 9 (1): 016108. <https://doi.org/10.1063/5.0236108> uses 20 PPW with 0.05 CFL. So you can use this combination for publishable data. If you only need to check quickly your parameters, you can use lower values

We tested our code on a Intel ® , Core™ i9-14900K, GPU NVIDIA RTX 4000 Ada Generation. High PPW and low CFL will struggle to run on a laptop, so it's recommended to use a PC with GPU.

1. Set **Desired focal pressure** to the pressure you want in free water.
2. Click **Run simulation**. It calibrates in water first, then runs through the skull.
3. When it finishes you get three views of the beam (skull in white, pressure in colour, a star at the focus), and a results panel with:
 - pressure at the focus, and how much got through the skull
 - **I_SPPA** and **I_SPTA** (intensity, in W/cm²)
 - **MI**, the mechanical index
4. Below that, set the pulsing (PRF, pulse duration, sonication time, cooling time) and click **Run thermal** to get the temperature rise and thermal dose.

If the focus temperature keeps rising during the cooling period, that is normal. Almost all the heat is made in the skull bone, and it takes tens of seconds to conduct inwards to the focus. Set a longer cooling time to see it come back down.

Tab 4: Protocol

This checks a whole planner CSV for overlapping heating.

1. Click **Load plan** (red button) and choose the CSV you saved from fus_planner.
2. Choose the skull model:
 - **Brain only** is fast and ignores the skull. Good for a first look and to see areas of overlap in heating.
 - **Per-spot acoustic** runs a full simulation for every spot through the skull. Accurate, but slow (especially if there are a lot of points to cover). Run the Simulation tab first, since it reuses that calibration.
3. Set the timing (how long each spot is on, how long to cool between spots).
4. Click **Run protocol**. You get the cumulative temperature, a map of where spots overlap, and the overlap volume in mm³.